

JGR Space Physics



RESEARCH ARTICLE

10.1029/2026JA035779

Key Points:

- Strong geoeffective bursts of FeXV 28.4 nm emission can occur even in the absence of recorded solar flares or after weak flares
- Statistical distributions of the amplitudes, occurrence rates, and growth durations of geoeffective FeXV 28.4 nm enhancements were obtained
- An empirical relationship between the total electron content response and FeXV 28.4 nm flux enhancements during the late phase of solar flares was obtained

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Citation:

Bekker, S. Z., O'Hare, A. N., Gcreatorex, H. J., Hayes, L. A., Janitzek, N., Kerr, G. S., et al. (2026). Statistical study of geoeffective Fe XV 28.4 nm emission bursts and their influence on ionospheric total electron content during the late phase of solar flares. *Journal of Geophysical Research: Space Physics*, 131, e2026JA035779. <https://doi.org/10.1029/2026JA035779>

Received 9 JUN 2026

Accepted 14 JUL 2026

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Statistical Study of Geoeffective Fe XV 28.4 nm Emission Bursts and Their Influence on Ionospheric Total Electron Content During the Late Phase of Solar Flares

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Abstract Recent studies have demonstrated that warm coronal emissions during the extreme ultraviolet late phase (ELP) of X-class solar flares can be sufficiently strong to also affect electron density in the Earth's ionosphere. The Fe XV 28.4 nm line was identified as one of the most geoeffective emissions produced during the ELP. In this study, we show that Fe XV 28.4 nm bursts with comparable amplitudes and growth rates may also follow significantly weaker flares. For this reason, a large statistical analysis of all Fe XV 28.4 nm irradiance enhancements observed from 2010 to 2014 was performed, resulting in the identification of 724 geoeffective events. The detected enhancements were associated with flare main phases (51%), long duration events (25%), late phases (15%), non-flare-related events (6%), or multiple-flare sequences (3%). The majority of ELP events (64%) were associated with M-class flares, while 8% of ELPs exhibited substantial Fe XV flux enhancements capable of producing ionospheric ionization comparable to that caused by the impulsive phase of X-class flares. In the second part of this study, the ionospheric total electron content response to the strongest and most isolated ELP events was evaluated using Fe XV emission measurements from several instruments covering the period 2010–2025. A total of 71 ELPs were identified. A strong positive relationship was obtained between the integrated Δ TEC enhancement and the integrated Fe XV 28.4 nm flux increase, with a correlation coefficient of $R = 0.7$. The empirical relationships provide a new quantitative tool for assessing the geoeffectiveness of warm coronal emissions and estimating the ionospheric response to ELPs.

1. Introduction

It is well known that the Earth's ionosphere is continuously exposed to solar radiation, which is its primary source of ionization. The geoeffectiveness of individual spectral lines and bands depends on a combination of factors, including the neutral composition of the medium, photon energy, and the ionization and absorption cross sections of specific particles (Bekker, 2018; Korsunskaja, 2019; Nishimoto et al., 2023; Solomon & Qian, 2005). For this reason, irradiance variations in some solar emissions do not produce a detectable ionospheric response, whereas enhancements in other spectral lines can lead to significant disturbances in ionospheric parameters. The latter often result in disruptions of modern communication and navigation systems (Habarulema et al., 2022; Sapunova et al., 2026; Yasyukevich et al., 2018), making the study of solar-terrestrial interactions an important problem in modern geophysics.

The most significant and impactful bursts of solar radiation are observed during solar flares, when substantial enhancements in ultraviolet and X-ray irradiance occur (Arnaut et al., 2023; Bekker et al., 2021; Leonovich et al., 2002; Lin et al., 2025; Mahmoudian et al., 2025; Mitra, 1974; Xiong et al., 2011). The power of a flare is commonly characterized by the magnitude of the soft X-ray emission it produces. Specifically, the flare class is determined from the emission in the 0.1–0.8 nm channel measured by the X-ray Sensor onboard the Geostationary Operational Environmental Satellites (GOES/XRS; Machol & Viereck, 2016). This classification is referred to as the “GOES class” of the flare. However, irradiance variations in other spectral ranges are not always proportional to the flare class (Gcreatorex et al., 2023). Therefore, to properly analyze the ionospheric response, it is necessary to consider the full solar spectrum, as far as allowed by modern space-based solar observatories and spectral models.

Writing – review & editing: S. Z. Bekker, A. N. O'Hare, H. J. Greatorex, L. A. Hayes, N. Janitzek, G. S. Kerr, R. O. Milligan, L. Qian, K. Watanabe

Due to the complex vertical structure of the Earth's neutral atmosphere, photons of different wavelengths penetrate to different altitudes before being absorbed. As a result, the spectral energy distribution of a flare determines the altitude redistribution of charged particles in the ionosphere (Bekker, 2025; Brasseur & Solomon, 2005; Nishimoto et al., 2023; Ryakhovsky et al., 2024; Watanabe et al., 2021). For example, X-ray radiation (< 10 nm) penetrates deepest and is the primary source of ionization in the lower ionosphere (D-region, 50–90 km), whereas emissions in the extreme ultraviolet (EUV, 10–121 nm) are mainly absorbed in the E- (90–140 km) and F-regions (> 140 km).

Long-term observations of solar emissions from instruments onboard satellites such as the Solar Dynamics Observatory (SDO; Pesnell et al., 2012), GOES, the Reuven Ramaty High Energy Solar Spectroscopic Imager (RHESSI; Lin et al., 2002), *Hinode* (Culhane et al., 2007; Kosugi et al., 2007), and Solar Orbiter (Müller et al., 2020) have enabled detailed investigation of the formation and evolution of solar flares, which are now well established to evolve through several distinct phases (Benz, 2017; Fletcher et al., 2011; Woods et al., 2011). The impulsive phase is characterized by the acceleration of non-thermal particles producing hard X-ray bursts and microwave emission. The associated energy deposition drives a rapid irradiance increase in cool chromospheric lines such as He II 30.4 nm, He I 58.4 nm, and C III 97.7 nm, which serve as the primary source of ionization for the upper and middle ionosphere during flares (Nishimoto et al., 2023; O'Hare et al., 2025). This is followed by the gradual phase, during which X-ray emission and hot coronal lines begin to decay, leading to enhanced ionization at lower altitudes (Bekker, 2025; Watanabe et al., 2021). The impulsive and gradual phases of a solar flare together are referred to as the main phase. Woods et al. (2011) and Woods (2014) demonstrated that warm coronal lines (e.g., Fe XV 28.4 nm, Fe XVI 33.5 nm, with formation temperatures of ~ 2 MK and ~ 2.7 MK, respectively) may exhibit a secondary peak minutes to hours after the main phase of a flare. This phenomenon is known as the extreme ultraviolet (EUV) late phase (ELP) and has been found to produce a second wave of ionization (Bekker et al., 2024; J. Liu et al., 2024).

Not all flares exhibit an EUV late phase. Various studies report different occurrence rates, but in general, higher-class flares are more likely to be accompanied by an ELP. Greatorex et al. (2026), using SDO's Atmospheric Imaging Assembly (SDO/AIA; Lemen et al., 2012) solar disk images in the Fe XVI 33.5 nm emission line, identified 467 ELP events among 5,335 isolated flares between 2010 and 2025, corresponding to an overall occurrence rate of 9%. The occurrence was higher for X-class (19%) and M-class flares (19%) compared to C-class flares (7%) and peaked in the M3.0–M5.9 range. Ormig et al. (2025) reported that approximately 10% of flares with GOES class $\geq C3.0$ exhibit an EUV late phase. Earlier studies, such as Woods (2014), found a 14% occurrence rate for flares of class C2.0 and above, with up to 50% for X-class flares, although this estimate was based on a smaller sample.

Despite occurring in a non-negligible fraction of flares, the influence of the EUV late phase on the ionosphere was only identified in 2024. Analyzing individual X-class flares, Bekker et al. (2024) and J. Liu et al. (2024) demonstrated that TEC exhibits a measurable increase in response to late phase enhancements in warm coronal lines such as Fe XV 28.4 nm and Fe XVI 33.5 nm. This finding highlights that the EUV late phase should now be accounted for in ionospheric modeling in response to solar flares. A statistical analysis of TEC measurements from 956 sunlit Global Navigation Satellite Systems (GNSS) stations and solar irradiance data from SDO's Multiple EUV Grating Spectrographs-A (SDO/MEGS-A) revealed that Fe XV 28.4 nm is the most geoeffective emission during the late phase (Bekker et al., 2024). The same study showed that the widely used Flare Irradiance Spectral Model-2 (FISM2; Chamberlin et al., 2020) currently does not reproduce ELP emissions as it was not designed to. This means that ionospheric models based on FISM2 cannot capture the associated ionospheric variations driven by flare ELP. Bekker et al. (2025) presented the first multi-event study of the ELP impact on TEC, deriving an empirical relationship between ΔTEC and the Fe XV 28.4 nm irradiance increase using 14 X-class flares with isolated late phases observed between 2010–2014 and 2017–2024.

Previous studies of ELP ionospheric effects have focused on X-class flares, which is not surprising since they are typically expected to exhibit the strongest Fe XV 28.4 nm enhancements. However, we report here that comparable Fe XV 28.4 nm flux increases can also follow significantly weaker flares, or even occur in the absence of any detected flare. Since increases in Fe XV 28.4 nm emission lead to enhanced ionization regardless of the underlying physical mechanism on the Sun, it is important not to limit the analysis to X-class flares but to consider all Fe XV 28.4 nm increases exceeding a certain threshold. Therefore, the first part of this study performs a statistical analysis of all geoeffective Fe XV 28.4 nm bursts (exceeding thresholds in amplitude and rise rate)

observed during the operation of MEGS-A (Woods et al., 2012) from April 2010 to May 2014. A total of 724 events were identified, including those associated with different flare phases and events not linked to any GOES-detected flare. Such an analysis is performed here for the first time, offering new insights into the timing, occurrence rate, and intensity of these events.

In the second part of this study, we focus on the strongest and most isolated Fe XV 28.4 nm bursts associated exclusively with the EUV late phase to evaluate their impact on TEC using GNSS data. The analysis period is extended using Fe XV 28.4 nm observations from the GOES-R (Eparvier et al., 2009) series (available since 2017). Although the average magnitude of Fe XV 28.4 nm enhancements occurring during the ELPs of X-class flares is larger, the much higher occurrence rate of C- and M-class flares allowed us to identify 71 suitable isolated events that produced measurable TEC increases. In contrast to earlier studies, which focused only on X-class flares, this work considers a substantially larger and more diverse sample of flares of various classes exhibiting late phase emission, for which the ionospheric response was assessed. Using this expanded data set, we derive a new empirical relationship between Fe XV 28.4 nm irradiance enhancement and the corresponding ionospheric response.

Section 2 of this paper presents the results of the statistical analysis of Fe XV 28.4 nm bursts from 2010 to 2014. Section 3 focuses on the ionospheric response to isolated Fe XV 28.4 nm enhancements and the derivation of the empirical relationship between the warm coronal emission increase and TEC dynamics. Section 4 provides discussion of the results and conclusions.

2. Statistical Analysis of Geoeffective Bursts in Fe XV 28.4 nm

2.1. Solar Observations

Solar EUV spectral irradiance measurements were obtained using the MEGS-A (6–37 nm) instrument onboard the SDO satellite. From April 2010 to May 2014, it monitored variations in solar radiation with high temporal resolution (10 s). To identify geoeffective enhancements in warm coronal lines, we analyzed measurements of the Fe XV 28.4 nm emission. In addition, another geoeffective warm coronal emission, the Fe XVI 33.5 nm line, was considered to compare their relative intensity and timing.

We also used measurements of the cool chromospheric/transition region He II 30.4 nm emission from MEGS-A, which dominates ionospheric ionization during the main phase. This emission was used as an indicator of background disturbances to assess the isolation of the Fe XV 28.4 nm bursts analyzed in this study. Although EUV measurements are available at 10 s cadence, a 1-min resolution was sufficient for the purposes of this work; therefore, Level 2B data with 1-min cadence were used to reduce computational cost.

To associate the analyzed events with specific solar flares, we generated a flare list of events during this time period from the Heliophysics Event Knowledgebase (HEK; Hurlburt et al., 2012) catalog of GOES flare events.

2.2. Selection Method

SDO/MEGS-A Level 2B spectral line data were searched for rapid Fe XV 28.4 nm irradiance enhancements. Prior to event detection, each Fe XV 28.4 nm time series was preprocessed: (a) non-finite and negative values were treated as non-physical and removed, (b) extended flatline segments (runs of ≥ 10 consecutive samples with zero differences) were excluded, and (c) a short centered rolling-median filter (window width: 5 samples; minimum 3 finite samples per window) was applied to suppress isolated spikes and noise while preserving the overall trend.

Bekker et al. (2025) demonstrated that if the Fe XV 28.4 nm flux increase over 1 hr (3600 s) exceeds $2 \times 10^{-6} \text{ W m}^{-2}$, a corresponding increase in ionospheric electron density can be reliably detected. Therefore, the values $\Delta F_{\text{thr}} = 2 \times 10^{-6} \text{ W m}^{-2}$ and $\Delta F_{\text{thr}}/\Delta t = 2 \times 10^{-6}/3600 \text{ W m}^{-2} \text{ s}^{-1}$ were used as function- and derivative-thresholds for selecting potentially geoeffective events.

The first set of candidate events was identified using a derivative-threshold approach. All time intervals where the derivative exceeded the fixed threshold were flagged as potential events. For each candidate interval, the event start time ($t_{\text{Fe XV start}}$) was defined as the moment when the flux derivative first exceeded $\Delta F_{\text{thr}}/\Delta t$, and the event peak time as the moment when it dropped below this threshold, corresponding to the end of the rapid-rise phase. To

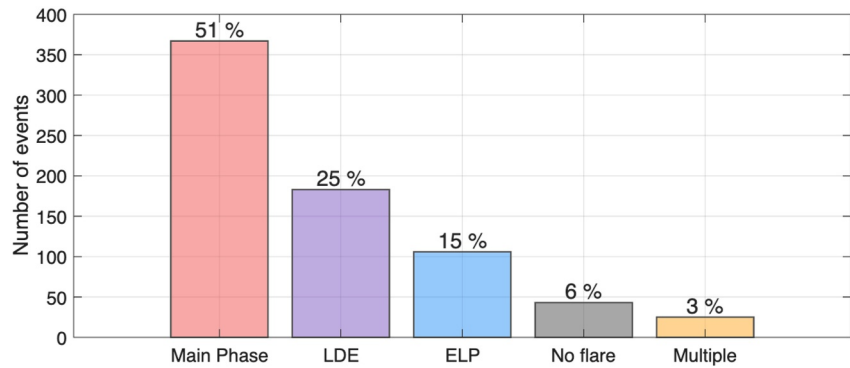


Figure 1. Number of geoeffective Fe XV events from April 2010 to May 2014 and their distribution by category described in Section 2.2.

estimate event amplitudes robustly with respect to small timing uncertainties, the start flux ($F_{\text{Fe XV start}}$) was taken as the minimum smoothed irradiance within ± 5 minutes of the event start time, while the peak flux ($F_{\text{Fe XV peak}}$) was taken as the maximum smoothed irradiance within ± 5 minutes of the event peak time. The event amplitude was defined as $\Delta F_{\text{Fe XV}} = F_{\text{Fe XV peak}} - F_{\text{Fe XV start}}$, and the rise duration as $\Delta t_{\text{Fe XV}} = t_{\text{Fe XV peak}} - t_{\text{Fe XV start}}$.

Final events were required to satisfy both an amplitude and a rate constraint:

$$\Delta F_{\text{Fe XV}} \geq \Delta F_{\text{thr}} \quad ; \quad \frac{\Delta F_{\text{Fe XV}}}{\Delta t_{\text{Fe XV}}} \geq \frac{\Delta F_{\text{thr}}}{\Delta t}$$

which prevents slowly varying increases from being classified as events. A total of 724 geoeffective bursts of the Fe XV 28.4 nm line were identified for the considered period. Lowering this amplitude threshold by 25% leads to an almost twofold increase in the number of events (1281 events), whereas raising the threshold by 25% results in a substantial decrease in the number of events (356 events). Nevertheless, we settled on a value of $\Delta F_{\text{thr}} = 2 \times 10^{-6} \text{ W m}^{-2}$, above which we can observe reliable dynamics of ionospheric electron density. All 724 events were classified into five categories according to their physical origin: “Main phase”, “Late phase” (ELP), “Long duration event” (LDE), “No flare”, and “Multiple”.

“Main phase” events were those for which the Fe XV 28.4 nm flux reached its maximum between the GOES flare start and end times. LDEs were defined as events that began during a flare and peaked after the flare end. ELP events were those that peaked after the flare end but were preceded by an initial Fe XV 28.4 nm peak during the main flare phase. These were manually classified as ELPs using the criteria outlined by Greatorex et al. (2026) and Woods (2014). “No flare” events were Fe XV 28.4 nm enhancements that satisfied the detection criteria but were not associated with any GOES-recorded flare (no solar flare was recorded within at least 6 hr prior to the Fe XV enhancement). The “Multiple” category includes 25 Fe XV 28.4 nm events occurring during sequences of multiple flares, sometimes originating from different active regions (recall that SDO/MEGS-A, and GOES/XRS are a disk-integrated detectors). Since this group represents a small fraction of the data set ($\sim 3\%$) and is difficult to uniquely associate with a specific flare, it was excluded from further analysis. The distribution of events across categories is shown in Figure 1. Figure 2 presents examples of the four main event types.

2.3. Results of the Statistical Analysis

2.3.1. Amplitude and Rise Duration of Fe XV 28.4 nm Bursts

To understand the frequency and significance of possible electron density variations in the ionosphere induced by Fe XV 28.4 nm enhancements, it was first necessary to analyze the distributions of these events in terms of their amplitude and rise duration. Figure 3 shows the distributions of Fe XV 28.4 nm flux increase amplitudes ($\Delta F_{\text{Fe XV}}$; left panels) and rise durations (right panels). Since stronger events are inherently much rarer, for all four

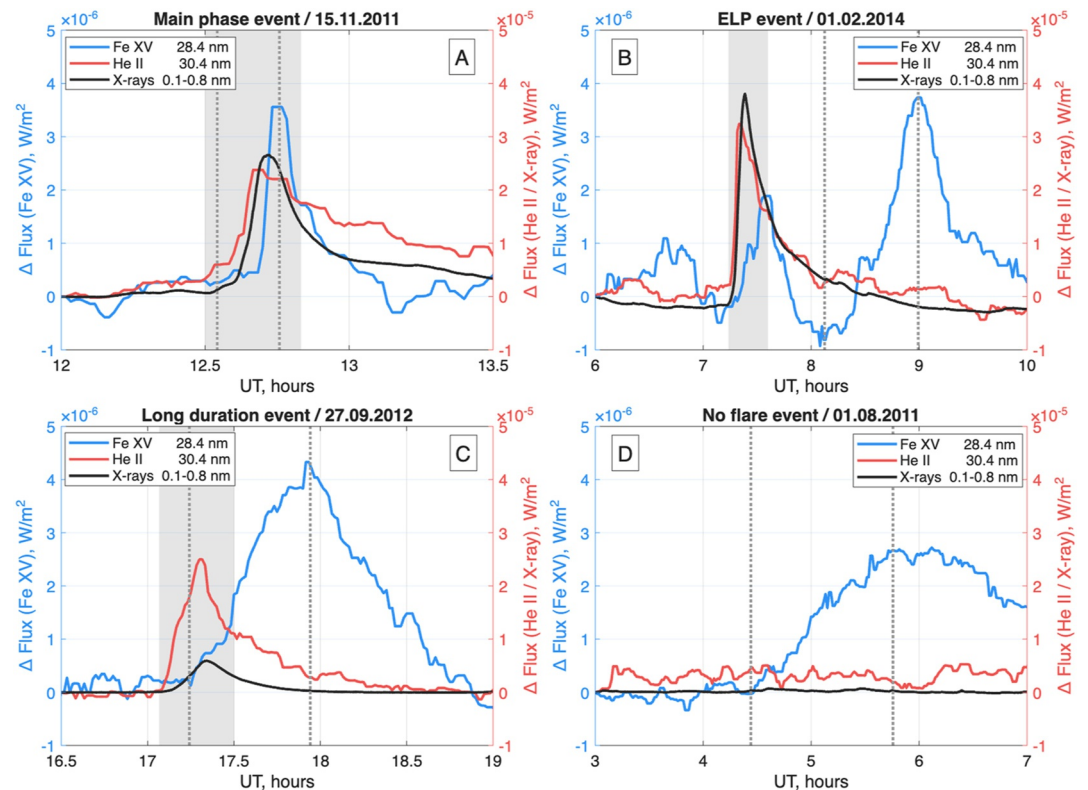


Figure 2. Examples of different categories of Fe XV 28.4 nm emission enhancement. The left vertical axis shows relative Fe XV irradiance (blue), while the right axis shows relative variations in He II 30.4 nm irradiance (red) and X-rays 0.1–0.8 nm (black). Panel (a) Main phase event (associated with M2.7, 15 November 2011, 12:43 UT). Panel (b) EUV late phase event (associated with M4.3, 1 February 2014, 07:23 UT). Panel (c) Long duration event (associated with C6.3, 27 September 2012, 17:20 UT); unlike the late phase, the rise in warm coronal emission begins almost simultaneously with the flare onset, and reaches a single maximum after the flare ends. Panel (d) No flare event (1 August 2011); no significant variations in He II or X-ray emission are observed prior to or during the event. Vertical dotted lines indicate the start and peak times of the Fe XV events, while semi-transparent gray shaded areas denote the flare intervals.

categories the largest fraction falls within the weakest considered amplitude range of $(2 \times 10^{-6} - 4 \times 10^{-6}) \text{ W m}^{-2}$, with the occurrence rate decreasing further as amplitude increases.

Panels A and B correspond to bursts occurring during the main phase of solar flares (red bars). In terms of amplitude, 70% of these events fall within the range $(2 \times 10^{-6} - 4 \times 10^{-6}) \text{ W m}^{-2}$. According to Bekker et al. (2025), such events can produce an ionospheric TEC increase of approximately ~ 0.2 TECU. A further 18% of events fall within $(4 \times 10^{-6} - 6 \times 10^{-6}) \text{ W m}^{-2}$, corresponding to a more significant contribution to the ionization of approximately ~ 0.3 TECU. Finally, in 12% of cases, the Fe XV 28.4 nm flux increase exceeds $6 \times 10^{-6} \text{ W m}^{-2}$, leading to even stronger electron density perturbations. The relative contribution of Fe XV 28.4 nm events to the total ionization is also determined by other geoeffective components of the solar spectrum, primarily the He II 30.4 nm emission. During main phase events, the average He II flux enhancement is the highest among all event groups, reaching $3.5 \times 10^{-5} \text{ W m}^{-2}$, thereby reducing the relative contribution of Fe XV to the overall ionization. The mean duration of the Fe XV 28.4 nm rise, defined as $D = t_{\text{Fe XV peak}} - t_{\text{Fe XV start}}$, is shortest, with an average value of 33 min.

Panels c and d present results for long duration events (purple bars), which are also characterized by a single-peaked Fe XV 28.4 nm profile, but with the peak occurring after the flare end time. Consequently, the rise duration is significantly longer (see Panel d), with the peak reached on average after 55 min from the onset of the increase. The amplitude distribution is shifted toward lower values (see Panel c) compared to main phase events: 84% of events fall within $\Delta F_{\text{Fe XV}} < 4 \times 10^{-6} \text{ W m}^{-2}$. Thus, the absolute contribution of warm coronal emission to the overall flare-driven ionization at the event peak is, on average, smaller for these events. However, its time-

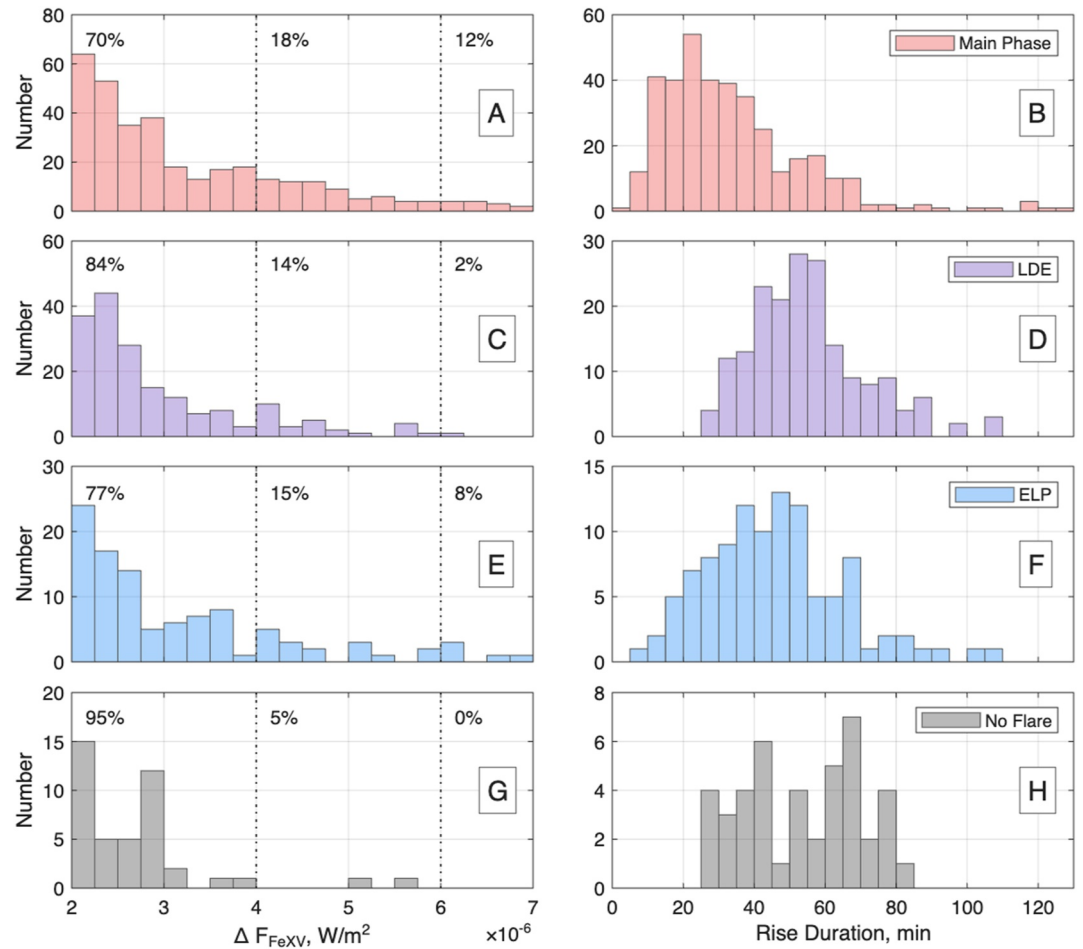


Figure 3. Distribution of Fe XV 28.4 nm flux increase amplitudes (a, c, e, g) and rise durations (b, d, f, h) for events during the main phase (a), (b), long duration events (c), (d), EUV late phase events (e), (f), and non-flare events (g), (h).

integrated contribution is expected to be larger, as these events last nearly twice as long as main phase events. The average He II flux change for this group is $1.0 \times 10^{-5} \text{ W m}^{-2}$.

Panels E and F show analogous distributions for EUV late phase events (blue bars). The mean Fe XV 28.4 nm rise duration during ELPs is 45 min (Panel f), which is shorter than for LDEs but longer than for main phase events. Compared to LDEs, a larger fraction of events exhibits higher amplitudes: 15% fall within $(4 \times 10^{-6} - 6 \times 10^{-6}) \text{ W m}^{-2}$, and 8% exceed $6 \times 10^{-6} \text{ W m}^{-2}$. It is important to note that, due to the varying geoeffectiveness of different solar emissions, the same magnitude of increase in irradiance flux can produce a different ionospheric response depending on the wavelength of the emission. According to Bekker et al. (2025), a $\Delta F_{\text{He II}}$ value of 10^{-5} W m^{-2} corresponds to a TEC increase of only $\sim 0.1 \text{ TECU}$, which is several times smaller than the ionospheric response potentially driven by the late phases considered in this work ($\Delta F_{\text{Fe XV}} \geq 2 \times 10^{-6} \text{ W m}^{-2}$). For ELP events, background variations of the He II 30.4 nm flux decrease to $\Delta F_{\text{He II}} = 9.6 \times 10^{-6} \text{ W m}^{-2}$, reflecting the more isolated nature of late phase events.

Finally, “No flare” events shown in Panels g and h (gray bars) are almost entirely concentrated in the lowest amplitude range, with an average rise duration of 53 min. Although these events do not exhibit large amplitudes or sharp increases, their presence is important, as they can produce electron density variations even in the absence of any solar flare. The average He II background variability for this group is only $8.3 \times 10^{-6} \text{ W m}^{-2}$.

The numerical parameters of Fe XV 28.4 nm bursts discussed in Section 2.3.1 are summarized in Table 1.

Table 1
Amplitude and Rise Duration Parameters of Fe XV 28.4 nm Bursts

	Main phase	LDE	ELP	No flare
Number of events	367	183	106	43
Mean $\Delta F_{\text{Fe XV}} \pm \sigma$ [W/m ²]	$(3.9 \pm 2.7) \times 10^{-6}$	$(3.0 \pm 1.1) \times 10^{-6}$	$(3.3 \pm 1.7) \times 10^{-6}$	$(2.7 \pm 0.7) \times 10^{-6}$
Median $\Delta F_{\text{Fe XV}} \pm \sigma$ [W/m ²]	$(2.9 \pm 1.7) \times 10^{-6}$	$(2.6 \pm 0.8) \times 10^{-6}$	$(2.7 \pm 1.1) \times 10^{-6}$	$(2.6 \pm 0.5) \times 10^{-6}$
Event fraction for $2 \times 10^{-6} \leq \Delta F_{\text{Fe XV}} < 4 \times 10^{-6}$	70%	84%	77%	95%
Event fraction for $4 \times 10^{-6} \leq \Delta F_{\text{Fe XV}} < 6 \times 10^{-6}$	18%	14%	15%	5%
Event fraction for $\Delta F_{\text{Fe XV}} \geq 6 \times 10^{-6}$ [%]	12%	2%	8%	0%
Mean D $\pm \sigma$ [min]	33 $\pm$ 20	55 $\pm$ 16	45 $\pm$ 19	53 $\pm$ 17
Median D $\pm \sigma$ [min]	29 $\pm$ 15	53 $\pm$ 13	44 $\pm$ 15	54 $\pm$ 15
Average $\Delta F_{\text{He II}}$ [W/m ²]	3.5×10^{-5}	1.0×10^{-5}	9.6×10^{-6}	8.3×10^{-6}

2.3.2. Distribution of Fe XV 28.4 nm Bursts by Solar Flare Class

Since long duration events are essentially part of the main phase of a solar flare, the “LDE” category was included in the “Main phase” group for further analysis.

For Fe XV 28.4 nm enhancements during the main phase, approximately equal numbers of events were associated with C- and M-class flares (about ~ 250 each), while the number of events corresponding to X-class flares was about five times lower. This is explained by the substantially lower overall occurrence rate of X-class flares. The final sample also included five Fe XV bursts ($\sim 1\%$) associated with the main phases of B-class flares. Despite their weak X-ray emission, the corresponding warm coronal emissions exceeded the geoeffectiveness thresholds.

For Fe XV 28.4 nm bursts occurring during the EUV late phase (i.e., ELP events), the majority of events (64%) are associated with M-class flares. This may be due to the balance between the occurrence rate of these flares and their ability to produce secondary emission. In contrast, based on the analysis of flares from 2010 to 2025 by Grotorex et al. (2026), only 28% of ELP events were associated with M-class flares. The lower fraction reported in that study may be due to the absence of a threshold on the minimum $\Delta F_{\text{Fe XV}}$, and thus the inclusion of a large number of weaker C-class flares, which constitute the majority of ELP events (68%) in their data set.

The numerical parameters describing the association between Fe XV 28.4 nm increases and solar flare classes, discussed in Section 2.3.2, are presented in Table 2.

2.3.3. Fe XV 28.4 nm Bursts Relative to Solar Flare Timing

The next aspect considered in the analysis was the relative timing between solar flare onset and peak times ($t_{\text{Flare start}}$ and $t_{\text{Flare peak}}$) and the corresponding timing of Fe XV 28.4 nm emission ($t_{\text{Fe XV start}}$ and $t_{\text{Fe XV peak}}$), shown in Figure 4. During the main phase of solar flares, the onset of Fe XV 28.4 nm flux enhancement approximately

coincides with the onset of X-ray emission. As a result, the probability density function of $\Delta t_{\text{start}} = t_{\text{Fe XV start}} - t_{\text{Flare start}}$ is nearly symmetric around zero (left panel, red bars). Despite the nearly simultaneous onset, the warm coronal Fe XV 28.4 nm emission increases more slowly than the X-ray flux and, on average, reaches its maximum 27 min after the flare peak (right panel, red bars), while the median delay of $\Delta t_{\text{peak}} = t_{\text{Fe XV peak}} - t_{\text{Flare peak}}$ is approximately 18 min.

For the EUV late phase, which corresponds to a secondary peak of warm coronal emissions, only positive values of Δt_{start} are observed, as expected (left panel, blue bars). The mean and median delays of the Fe XV 28.4 nm flux enhancement onset relative to the flare start are 59 and 45 min, respectively.

Table 2
Distribution of Fe XV 28.4 nm Bursts by Solar Flare Class

	Main phase	ELP
Number of events	550	106
Number and fraction of B-class flares	5/1%	0/0%
Number and fraction of C-class flares	248/45%	30/28%
Number and fraction of M-class flares	253/46%	68/64%
Number and fraction of X-class flares	44/8%	8/8%

Note. The “LDE” category was included in the “Main phase” group.

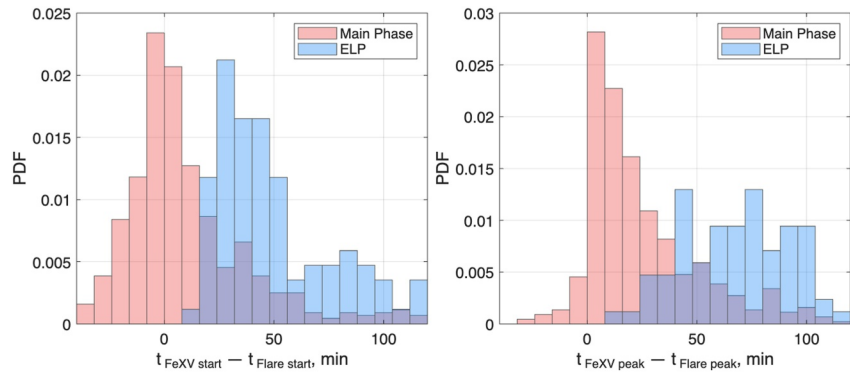


Figure 4. Left panel: Probability density function of the delay in the onset of Fe XV 28.4 nm emission relative to the flare start time. Right panel: Probability density function of the delay of the Fe XV 28.4 nm emission peak relative to the flare peak. The sample sizes are $N = 550$ for the main phase events and $N = 106$ for the ELP events.

The mean and median delays of the ELP peak relative to the flare peak are 89 and 79 min, respectively (right panel, blue bars). Gretores et al. (2026) reported very similar values of 88 and 82 min, respectively.

The temporal parameters of Fe XV 28.4 nm bursts relative to solar flare timing, discussed in Section 2.3.3, are summarized in Table 3.

2.3.4. Temporal and Amplitude Relationship Between Warm Coronal Fe XV 28.4 nm and Fe XVI 33.5 nm Emission Enhancements

In addition to the warm coronal Fe XV 28.4 nm line, the SDO/MEGS-A instrument also measures another geoeffective warm coronal emission, Fe XVI 33.5 nm. Previous studies (Bekker et al., 2024, 2025) demonstrated that Fe XV 28.4 nm correlates better with variations in ionospheric electron density during the late phase of solar flares, suggesting that this line contributes more strongly to ionization. Nevertheless, Fe XVI 33.5 nm remains an important source of ionization in the F-region (Watanabe et al., 2021). Therefore, the timings and relative amplitudes of these two emissions were analyzed.

The left panel of Figure 5 shows the difference between the peak times of Fe XV 28.4 nm and Fe XVI 33.5 nm emissions. During the main phase, the two lines evolve almost synchronously, with the Fe XV 28.4 nm peak occurring on average only 2 min later. During the EUV late phase, however, the Fe XVI 33.5 nm flux reaches its maximum on average 7 min earlier than the Fe XV 28.4 nm flux.

The right panel of Figure 5 presents the difference between the flux enhancements of these two warm coronal emissions, $\Delta F_{\text{Fe XV}} - \Delta F_{\text{Fe XVI}}$. It was found that the 33.5 nm line exhibits larger flux increases during the main phase of solar flares, whereas during the EUV late phase they are comparable, with $\Delta F_{\text{Fe XV}}$, on average, marginally larger than $\Delta F_{\text{Fe XVI}}$.

Despite the potentially significant contribution of Fe XVI 33.5 nm to ionospheric ionization during different flare phases, absolute measurements of this line are unfortunately unavailable after May 2014, when the SDO/MEGS-A instrument ceased operation. Therefore, the analysis of the ionospheric response to warm coronal emission bursts presented in Section 3 is based only on the dynamics of the Fe XV 28.4 nm flux, which has previously demonstrated a strong correlation with electron density variations during the late phases of X-class flares (Bekker et al., 2025).

The relative timing and amplitudes of the warm coronal lines discussed in Section 2.3.4 are summarized in Table 4.

3. Ionospheric Response to the EUV Late Phase

Since the ionospheric effects of warm coronal emissions cannot be isolated from X-rays and other EUV emission during the main phase of a solar flare,

Table 3
Temporal Parameters of Fe XV 28.4 nm Bursts Relative to Solar Flare Timing

	Main phase	ELP
Mean $\Delta t_{\text{start}} \pm \sigma$ [min]	10 ± 34	59 ± 43
Median $\Delta t_{\text{start}} \pm \sigma$ [min]	3 ± 23	45 ± 23
Mean $\Delta t_{\text{peak}} \pm \sigma$ [min]	27 ± 35	89 ± 47
Median $\Delta t_{\text{peak}} \pm \sigma$ [min]	18 ± 24	79 ± 36

Note. The “LDE” category was included in the “Main phase” group.

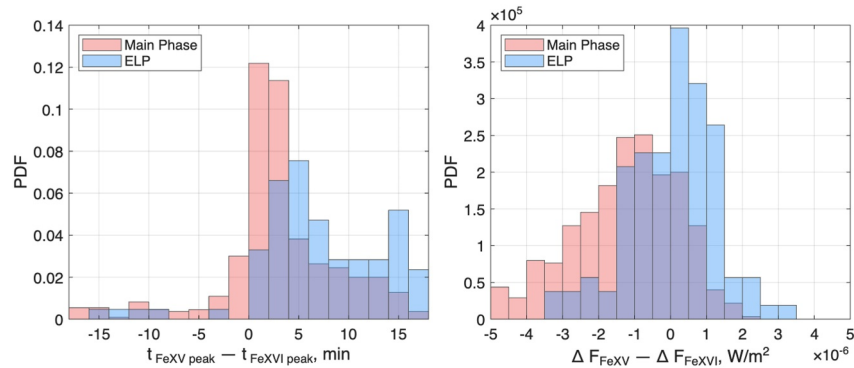


Figure 5. Left panel: Probability density function of the difference between the peak times of Fe XV 28.4 nm and Fe XVI 33.5 nm emissions. Right panel: Probability density function of the difference between the amplitudes of flux enhancements of Fe XV 28.4 nm and Fe XVI 33.5 nm emissions. The sample sizes are $N = 550$ for the main phase events and $N = 106$ for the ELP events.

only the impact of Fe XV 28.4 nm emission enhancements during the ELPs is considered in Section 3. Section 2 demonstrates that a large number of geoeffective Fe XV 28.4 nm emission bursts during ELPs are associated with relatively weak C- and M-class solar flares. This finding has allowed us to significantly expand, compared to previous studies, the set of potential impactful ELPs and to quantitatively assess the influence of the late phase on the Earth's ionosphere based on a more statistically robust data set.

3.1. Experimental Data

The phase and code delays of GNSS signals allow the estimation of the temporal dynamics of TEC in the ionosphere driven by variations in solar irradiance. To analyze the ionospheric response, as in previous studies, we used experimental measurements from the SOPAC (Scripps Orbit and Permanent Array Center) GNSS network. This network consists of more than 3000 stations globally, each of which can simultaneously track 4–5 satellites. The data cadence varies between stations and typically is 15 or 30 s.

Since photoionization occurs only on the sunlit portion of the ionosphere, for each event we considered only stations for which the solar elevation angle exceeded 30° . The latitude range was restricted to $\pm 55^\circ$ to avoid polar ionospheric effects. Due to the non-uniform distribution of GNSS stations, the number of stations used in the analysis could vary by orders of magnitude depending on the local time of the event.

To estimate the TEC change (ΔTEC) caused by the radiation increase, the relative TEC measurements were detrended to remove the parabolic trend associated with satellite trajectory. The detrending method used in this paper allows us to isolate the ionospheric response exclusively to the late phase Fe XV 28.4 nm enhancement, since it subtracts the background immediately preceding the onset of the late phase. As a result, the ΔTEC curves contain no contribution from residual ionization associated with the flare's main phase. The detrending procedure is described in detail in previous studies (e.g., Bekker et al., 2024) and enables an accurate estimation of ΔTEC over sunlit GNSS stations. However, it can only be applied over relatively short time intervals; therefore, the ionospheric response could be analyzed only for events shorter than 1.5 hr.

Table 4
Relative Timing and Amplitudes of Warm Coronal Fe XV 28.4 nm and Fe XVI 33.5 nm Emissions

	Main phase	ELP
Mean ($t_{\text{Fe XV peak}} - t_{\text{Fe XVI peak}} \pm \sigma$ [min])	2 ± 8	7 ± 9
Median ($t_{\text{Fe XV peak}} - t_{\text{Fe XVI peak}} \pm \sigma$ [min])	2 ± 5	6 ± 7
Mean ($\Delta F_{\text{Fe XV}} - \Delta F_{\text{Fe XVI}} \pm \sigma$ [W/m^2])	$(-1.2 \pm 1.5) \times 10^{-6}$	$(0.1 \pm 1.2) \times 10^{-6}$
Median ($\Delta F_{\text{Fe XV}} - \Delta F_{\text{Fe XVI}} \pm \sigma$ [W/m^2])	$(-1.1 \pm 1.2) \times 10^{-6}$	$(0.2 \pm 1.0) \times 10^{-6}$

Note. The “LDE” category was included in the “Main phase” group.

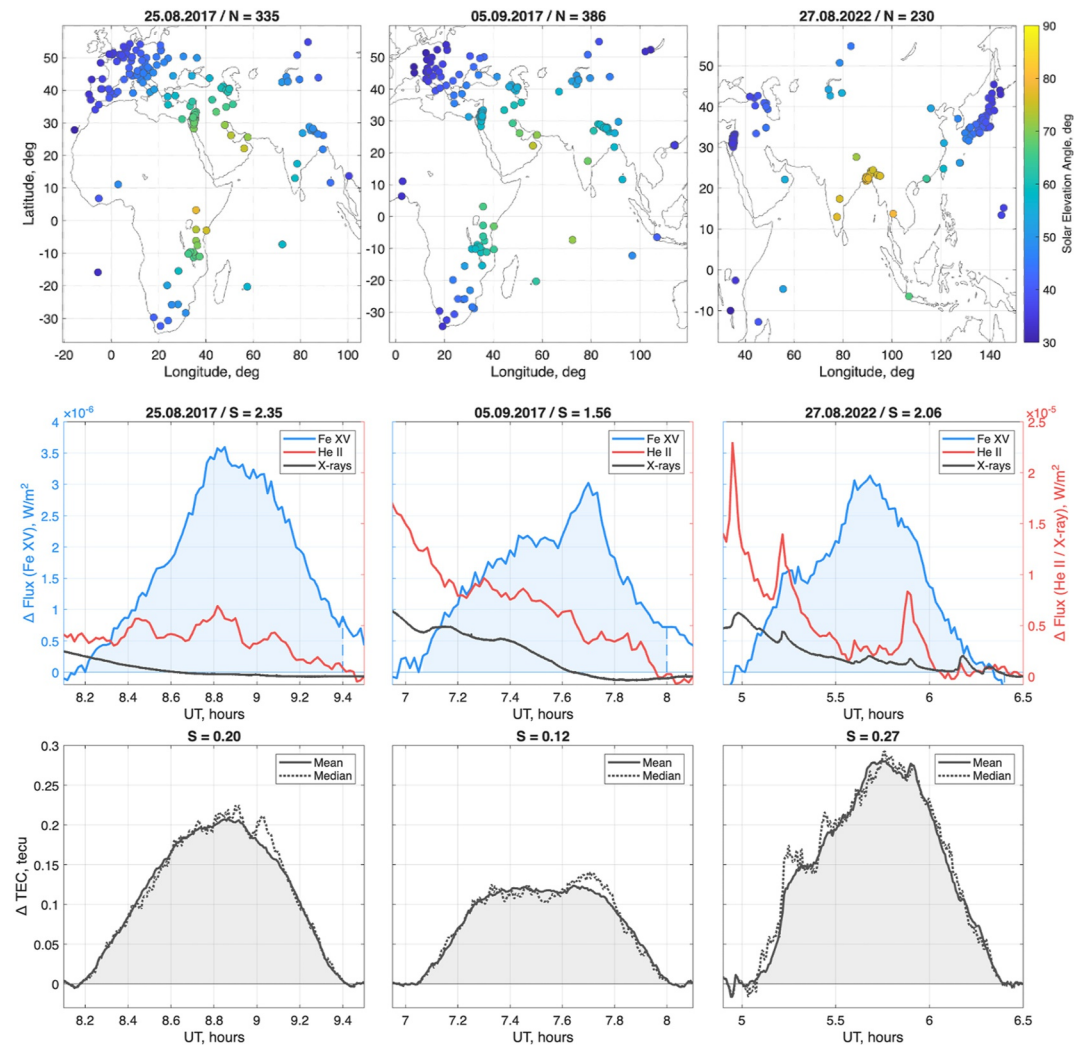


Figure 6. Top panels: Locations of global navigation satellite systems (GNSS) stations whose data were used to calculate TEC dynamics during the late phases of flares on 25 August 2017 (07:27 UT, C7.9), 5 September 2017 (06:40 UT, M5.4), and 27 August 2022 (02:40 UT, M4.8). Middle panels: Relative irradiance of Fe XV 28.4 nm (EUVS), He II 30.4 nm (EUVS), and X-rays (0.1–0.8 nm; GOES/XRS) during the corresponding late phases. Bottom panels: Average Δ TEC dynamics measured with GNSS during the corresponding late phases.

In this section, the analysis is not limited to the operational period of the MEGS-A instrument, since starting from 2017, measurements of Fe XV 28.4 nm and He II 30.4 nm irradiance from the GOES-R Extreme Ultraviolet Sensor (GOES-R/EUVS; Eparvier et al., 2009; Snow et al., 2009) became available with 1-min cadence. Since data from both instruments are calibrated to absolute irradiance values, they were combined and treated as a single data set. Thus, during the periods 2010–2014 and 2017–2025, we identified 71 isolated ELPs for the ionospheric response analysis, with amplitudes $\Delta F_{\text{Fe XV}} \geq 3 \times 10^{-6} \text{ W m}^{-2}$ and durations not exceeding 90 min.

3.2. Estimation of Total Electron Content Enhancements During the Late Phase

Figure 6 shows an example of estimating the TEC dynamics during three ELPs associated with C7.9 (25 August 2017, 07:27 UT, left), M5.4 (5 September 2017, 06:40 UT, center), and M4.8 (27 August 2022, 02:40 UT, right) flares. The upper panels of Figure 6 show the GNSS stations that were sunlit during the late phase and whose data were used to calculate Δ TEC. The numbers of stations were 335, 386, and 230 for the three events, respectively. The color indicates the solar elevation angle at each station location at the time of the ELP peak.

The ΔTEC variations obtained at each station were normalized by the square root of the sine of the solar elevation angle (Bekker et al., 2025; Ieda et al., 2014) to ensure a consistent averaging of data from differently illuminated locations.

The middle panels of Figure 6 show the relative irradiance of Fe XV 28.4 nm emission (blue; left y-axis), He II 30.4 nm emission (red; right y-axis), and X-ray emission in the 0.1–0.8 nm range (black; right y-axis). Despite the relatively weak flare classes, the Fe XV 28.4 nm flux enhancement during the late phase reaches significant values of $(3.0\text{--}3.5) \times 10^{-6} \text{ W m}^{-2}$, comparable to those observed even after strong X-class flares. The emission variations are shown for the ELP time intervals; therefore, only the decaying post-flare dynamics of the chromospheric He II 30.4 nm and X-ray emissions are observed. The variability of these emissions differs significantly between the three events, with the smallest changes observed in the left panel and the largest in the right panel.

The lower panels of Figure 6 present the station-averaged ΔTEC variations during the late phases, corrected for local illumination conditions. The resulting ΔTEC curves closely follow the temporal evolution of the Fe XV 28.4 nm flux and exhibit good timing agreement. The largest TEC enhancement is observed for the late phase of the M4.8 flare on 27 August 2022, despite the fact that the Fe XV 28.4 nm-driven ionizing flux for this event is lower than for the C7.9 flare on 25 August 2017. It is unlikely that the larger ionospheric response to the ELP observed on 27 August 2022 resulted from the higher background solar activity relative to 2017, since an already ionized ionosphere should be less responsive to further ionization. A significant latitudinal effect is also not expected, as the mean latitude of the GNSS stations used in the analysis was nearly identical for the two events. The most likely explanation is additional ionization associated with the second warm coronal emission, Fe XVI 33.5 nm, which was not measured for these events. Another contributing factor may be the observed intermittent bursts in He II 30.4 nm emission in the right panel, which can also contribute to ionization during the late phase. In contrast, in the left panel, both He II and X-ray variations are very small and can be considered background level.

3.3. Empirical Relationship Between TEC and Fe XV 28.4 nm Emission Enhancements

In a previous study (Bekker et al., 2025), which presented the first multi-event analysis of the impact of late phase warm coronal bursts on ionospheric electron density, the empirical relationship between $\Delta F_{\text{Fe XV}}$ and ΔTEC was derived using 14 X-class flares. Despite the limited sample size, a strong linear correlation between the parameters was obtained, with a Pearson correlation coefficient of $R_p = 0.84$.

Since ELPs cannot be completely isolated from other geoeffective spectral emissions, the resulting ΔTEC variations may be locally distorted, causing the electron density peak to occur at a different time than the Fe XV 28.4 nm irradiance peak. In such cases, comparing solar and ionospheric behavior solely using their maxima is not appropriate. Moreover, because both the irradiance enhancement and the associated ionospheric ionization evolve much more gradually during the late phase than during the impulsive phase, it is more physically meaningful to consider the integrated effect of warm coronal emissions rather than their peak values. This is further supported by X. Liu et al. (2024), who showed that sustained ELP irradiance can have an important cumulative effect on ionospheric electron density. For this reason, we analyzed not the maxima of the $\Delta F_{\text{Fe XV}}$ and ΔTEC functions, but the areas under the corresponding curves, $S(\Delta\text{Flux})$ and $S(\Delta\text{TEC})$, shown by the semi-transparent blue and gray shaded areas in the middle and lower panels of Figure 6, respectively. To avoid an artificial correlation caused by the common dependence of both parameters on event duration, both quantities were normalized by the ELP duration.

As a result of the analysis of TEC dynamics during isolated ELPs between 2010 and 2025, 71 pairs of $S(\Delta\text{Flux})$ and $S(\Delta\text{TEC})$ values were obtained. The relationship between the duration-normalized values is shown in the left panel of Figure 7. The data set includes ELPs for 9 C-class flares (13%), 55 M-class flares (77%), and 7 X-class flares (10%), shown in green, orange, and purple, respectively. An increase in TEC proportional to the integrated Fe XV 28.4 nm flux is observed.

Since variations in He II 30.4 nm emission can distort the magnitude of $S(\Delta\text{TEC})$, and the contribution of these variations to ionization cannot be reliably quantified, the next step involved filtering the events and excluding those for which He II variations during the ELP time interval exceeded 10^{-5} W m^{-2} . This value was chosen as a threshold based on previous estimates (Bekker et al., 2025), which showed that the geoeffectiveness of such a He II 30.4 nm flux change is relatively low and comparable to a Fe XV 28.4 nm flux change of approximately 10^{-6} W m^{-2} .

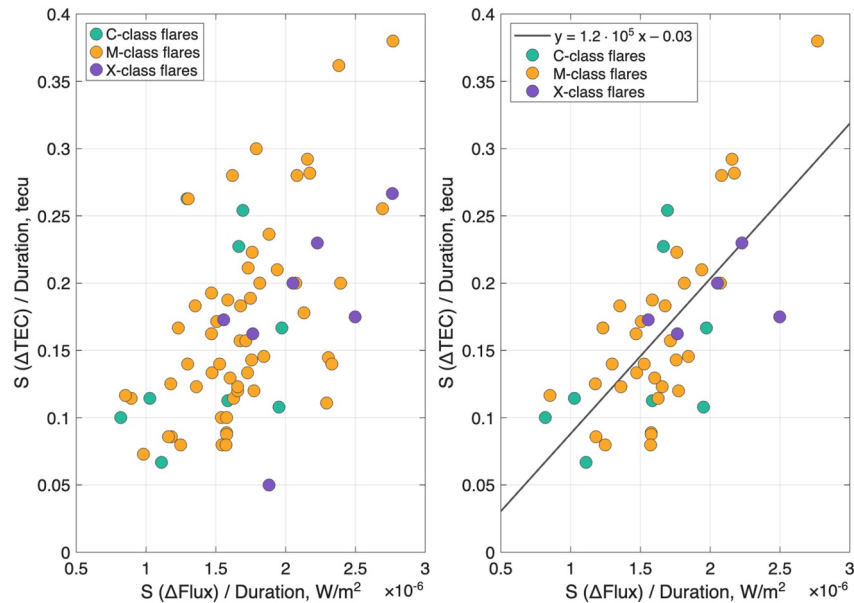


Figure 7. Dependence of the duration-normalized integrated ΔTEC enhancement on the duration-normalized integrated increase in irradiance in the Fe XV 28.4 nm. Left panel: Results for 71 EUV late phases; correlation coefficient for the linear fit is $R_p = 0.55$. Right panel: Results for 45 EUV late phases with no significant variations in He II 30.4 nm flux; correlation coefficient for the linear fit is $R_p = 0.70$.

After applying this selection criterion, 45 events remained (8 C-class flares, 32 M-class flares, and 5 X-class flares), shown as colored circles in the right panel of Figure 7. As seen in the figure, the filtering substantially reduces the scatter of the data points.

Table 5 presents the Spearman rank correlation coefficient (R_s), the Pearson correlation coefficient (R_p), and the root mean square error (RMSE) for the linear fit for both data sets. The Spearman correlation coefficients indicate a moderate relationship between the parameters in the original data set ($R_s = 0.54$) and a stronger correlation after filtering events with high He II 30.4 nm variability ($R_s = 0.65$). The Pearson correlation coefficient for the filtered data set is higher ($R_p = 0.70$), corresponding to a strong positive linear correlation between the integrated Fe XV 28.4 nm flux enhancement and the integrated TEC increase. The resulting linear relationship is described by the function $y = 1.2 \times 10^5 x - 0.03$, with a RMSE = 0.05. This line passes close to the point (0,0) (within the RMSE), as expected, since no increase in electron density should be observed in the absence of a driving ionization source.

Although the correlation coefficients are lower than those obtained previously for a smaller sample of X-class flares ($R_p = 0.84$; Bekker et al., 2025), they still indicate a robust relationship between the ELP's irradiance enhancement and the increase in the number of free electrons in the Earth's ionosphere. The empirical relationship obtained here provides important and reliable information about the geoeffectiveness of Fe XV 28.4 nm emission, which had previously been estimated using only a very limited sample of X-class flares and therefore remained more approximate. This relationship can be used as an empirical tool for preliminary estimation of the integrated ionospheric response to the EUV late phase from a given enhancement in Fe XV 28.4 nm irradiance.

4. Conclusions

The first studies devoted to enhanced ionization caused by late phase warm coronal emissions were based on a very limited number of X-class flares with pronounced late phases, and attempted to establish a relationship between the observed irradiance variations and the corresponding ionospheric response. In the present study, however, it has been shown that strong geoeffective Fe XV 28.4 nm bursts do not necessarily follow powerful solar flares, but may also occur after C- and M-class flares or even without any associated X-ray

Table 5
Statistical Correlation Metrics for Original and Filtered Event Data Sets

	Original data set (71 events)	Filtered Data set (45 events)
R_s	0.54	0.65
R_p	0.55	0.70
RMSE [W/m ²]	0.06	0.05

enhancement. Indeed, the correlation coefficient between flare class and Fe XV 28.4 nm irradiance enhancement during the late phase is only $R = 0.3$, indicating a rather weak relationship between these parameters.

The analysis of potentially geoeffective bursts of the warm coronal Fe XV 28.4 nm emission made it possible to determine their distributions in terms of amplitude and rise duration and to obtain new insights into these phenomena. In absolute terms, the late phase TEC enhancement is typically a few tenths of a TECU, comparable to the ionospheric response to an M-class or moderate X-class flare main phase, but well below that of large X-class flares. Its importance lies not in its magnitude but in the fact that it is a prolonged, additional source of ionization that current FISM2 based models do not capture. In particular, the Fe XV 28.4 nm irradiance enhancement during the ELP was found to be approximately 10% stronger and 22% faster than during long duration events, which are closely associated with the main flare phase. Approximately 8% of isolated events exhibited critical Fe XV 28.4 nm flux enhancements capable of producing substantial increases in F-region ionization. It was also found that the majority of isolated Fe XV 28.4 nm variations are associated with late phases of M-class flares (64%). This distribution may be explained by the combination of the high occurrence rate of M-class flares and their sufficient energy to drive secondary emission.

The statistical study demonstrated the importance of accounting for and measuring the second geoeffective warm coronal emission, Fe XVI 33.5 nm. Although this line previously showed a weaker correlation with the ionospheric response than Fe XV 28.4 nm, the analysis presented here indicates that its flux enhancement is comparable in magnitude to that of Fe XV 28.4 nm. Moreover, during the main phase, the mean Fe XVI 33.5 nm flux increase even exceeds that of Fe XV 28.4 nm. As the absolute Fe XVI 33.5 nm measurements have not been available since 2014, its direct ionospheric contribution remains unquantified. Its magnitude alone, however, indicates that future warm coronal monitoring should include this line.

An analysis of warm coronal emission bursts independent of flare class allowed us to identify a much larger number of geoeffective events than previously known and to conduct a large-scale study of their impact on ionospheric TEC. Using 45 selected strong isolated late phases, a new refined empirical relationship was derived between the duration-normalized integrated Δ TEC enhancement and the integrated Fe XV 28.4 nm flux increase. The Pearson correlation coefficient reached $R_p = 0.70$ for the linear fit, while the Spearman rank correlation was $R_s = 0.65$, indicating a strong correlation between the parameters. This relationship, obtained from a substantially larger and more diverse sample spanning C-, M-, and X-class flares, is consistent with that derived previously from a small set of X-class flares (Bekker et al., 2025), demonstrating that the effect of warm coronal emissions on TEC is robust beyond the strongest events.

Since the main phase of a solar flare is accompanied by irradiance enhancements in many spectral lines and continua, it is impossible to isolate their individual contributions to ionospheric ionization using experimental methods alone. The importance of detecting the ionospheric effect of the late phase also lies in the fact that it allows us to observe the isolated impact of warm coronal emissions on TEC. As a result, a combined analysis of the irradiance dynamics and the associated TEC enhancement makes it possible to empirically estimate the geoeffectiveness of warm coronal radiation.

The derived relationship demonstrates a robust connection between observed Fe XV 28.4 nm bursts and the ionospheric TEC response. Therefore, it represents a unique empirical tool both for estimating the TEC response to the late phase of solar flares and for assessing the contribution of warm coronal emissions during the main flare phase, when the ionization by different sources cannot be separated.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

No new data were generated as part of this research. Data from the SOPAC network are available at <http://sopac-old.ucsd.edu/>. The GOES/XRS data are available at https://www.ncei.noaa.gov/data/goes-space-environment-monitor/access/science/xrs/goes15/gxrs-l2-irrad_science/. The GOES-R/EUVS and SDO data used in this study are available at <https://data.ngdc.noaa.gov/platforms/solar-space-observing-satellites/goes/goes16/l2/> and https://lasp.colorado.edu/eve/data_access/eve_data/products/level2b/, respectively. This work utilized the open-source

Python package SunPy to obtain solar flare event lists and associated metadata (The SunPy Community et al., 2020).

Acknowledgments

S.Z.B., A.N.O., and R.O.M. would like to thank the European Office of Aerospace Research and Development (FA8655-22-1-7044-P00001) for supporting this research. H.J.G. and R.O.M. would like to thank the UK's Science & Technology Facilities Council (ST/X000923/1). G.S.K. acknowledges support from a Royal Society University Research Fellowship (URF\R1\251160). L.A.H. is supported by a Royal Society-Research Ireland University Research Fellowship (URF\R1\241775). This research was also supported by the International Space Science Institute (ISSI) in Bern, through International Team project #24–618.

References

- Arnaut, F., Kolarski, A., Srećković, V. A., & Mijić, Z. (2023). Ionospheric response on solar flares through machine learning modeling. *Universe*, 9(11), 474. <https://doi.org/10.3390/universe9110474>
- Bekker, S. (2025). Assessment of vertical redistribution of electron density in ionosphere during an X-Class solar flare using GNSS data. *Atmosphere*, 16(7), 825. <https://doi.org/10.3390/atmos16070825>
- Bekker, S. (2018). Analysis of electron density calculations using deterministic-probabilistic model of the ionospheric D-region. *Solar-Terrestrial Physics*, 4(3), 67–75. <https://doi.org/10.12737/stp-43201809>
- Bekker, S., Milligan, R. O., & Ryakhovsky, I. A. (2024). The influence of different phases of a solar flare on changes in the total electron content in the Earth's ionosphere. *The Astrophysical Journal*, 971(2), 188. <https://doi.org/10.3847/1538-4357/ad631d>
- Bekker, S. Z., Milligan, R. O., & Ryakhovsky, I. A. (2025). Response of ionospheric total electron content to the impulsive and late phases of X-class solar flares with various center-to-limb locations. *JGR*, 130(10), e2025JA034281. <https://doi.org/10.1029/2025JA034281>
- Bekker, S. Z., Ryakhovsky, I. A., & Korsunskaya, J. A. (2021). Modeling of the lower ionosphere during solar X-ray flares of different classes. *JGR*, 126(2), e2020JA028767. <https://doi.org/10.1029/2020JA028767>
- Benz, A. O. (2017). Flare observations. *Living Reviews in Solar Physics*, 4(2), 2. <https://doi.org/10.1007/s41116-016-0004-3>
- Brasseur, G. P., & Solomon, S. (2005). *Aeronomy of the middle atmosphere: Chemistry and physics of the stratosphere and mesosphere*. Cham: Springer.
- Chamberlin, P. C., Eparvier, F. G., Knoer, V., Leise, H., Pankratz, A., Snow, M., et al. (2020). The flare irradiance spectral model-version 2 (FISM2). *SpWea*, 18(12), e2020SW002588. <https://doi.org/10.1029/2020sw002588>
- Culhane, J. L., Harra, L. K., James, A. M., Al-Janabi, K., Bradley, L. J., Chaudry, R. A., et al. (2007). The EUV imaging spectrometer for Hinode. *Solar Physics*, 243(1), 19–61. <https://doi.org/10.1007/s01007-007-0293-1>
- Eparvier, F. G., Crotser, D., Jones, A. R., McClintock, W. E., Snow, M., & Woods, T. N. (2009). The extreme ultraviolet sensor (EUVS) for GOES-R. In S. Fineschi & J. A. Fennelly (Eds.), *Solar physics and space weather instrumentation iii* (Vol. 7438), 743804. SPIE Proceedings. <https://doi.org/10.1117/12.826445>
- Fletcher, L., Dennis, B. R., Hudson, H. S., Krucker, S., Phillips, K., Veronig, A., et al. (2011). An observational overview of solar flares. *Space Science Reviews*, 159(1–4), 19–106. <https://doi.org/10.1007/s11214-010-9701-8>
- Greatorex, H. J., Milligan, R. O., & Chamberlin, P. C. (2023). Observational analysis of Ly α emission in equivalent-magnitude solar flares. *The Astrophysical Journal*, 954(2), 120. <https://doi.org/10.3847/1538-4357/acea7f>
- Greatorex, H. J., O'Hare, A. N., Bekker, S., Campbell, R. J., Keane, D. C., & Milligan, R. O. (2026). The EUV late-phase: Statistical results from 15 years of solar dynamics observatory observations. *Solar Physics*, 301(6), 81. <https://doi.org/10.1007/s11207-026-02673-7>
- Habarulema, J. B., Tshisaphungo, M., Katamzi-Joseph, Z. T., Matamba, T. M., & Nndanganeni, R. (2022). Ionospheric response to the M- and X-Class solar flares of 28 October 2021 over the African sector. *Space Weather*, 20(8), e2022SW003104. <https://doi.org/10.1029/2022SW003104>
- Hurlburt, N., Cheung, M., Schrijver, C., Chang, L., Freeland, S., Green, S., et al. (2012). Heliophysics event knowledgebase for the Solar Dynamics Observatory (SDO) and beyond. In *The solar dynamics observatory* (pp. 67–78). Springer.
- Ieda, A., Oyama, S., Vanhamäki, H., Fujii, R., Nakamizo, A., Amm, O., et al. (2014). Approximate forms of daytime ionospheric conductance. *Journal of Geophysical Research (Space Physics)*, 119(12), 10397–10415. <https://doi.org/10.1002/2014JA020665>
- Korsunskaja, Y. (2019). Heuristic model of solar X-ray spectrum according to satellite data for geophysical applications. *Solar-Terrestrial Physics*, 5(3), 75–86. <https://doi.org/10.12737/stp-53201909>
- Kosugi, T., Matsuzaki, K., Sakao, T., Shimizu, T., Sone, Y., Tachikawa, S., et al. (2007). The Hinode (Solar-B) Mission: An overview. *Solar Physics*, 243(1), 3–17. <https://doi.org/10.1007/s11207-007-9014-6>
- Lemen, J. R., Title, A. M., Akin, D. J., Boerner, P. F., Chou, C., Drake, J. F., et al. (2012). The Atmospheric Imaging Assembly (AIA) on the solar dynamics observatory (SDO). *Solar Physics*, 275(1–2), 17–40. <https://doi.org/10.1007/s11207-011-9776-8>
- Leonovich, L. A., Afraimovich, E. L., Romanova, E. B., & Tschilil, A. V. (2002). Estimating the contribution from different ionospheric regions to the TEC response to the solar flares using data from the international GPS network. *Annales Geophysicae*, 20(12), 1935–1941. <https://doi.org/10.5194/angeo-20-1935-2002>
- Lin, R. P., Dennis, B. R., Hurford, G. J., Smith, D. M., Zehnder, A., Harvey, P. R., et al. (2002). The reuven ramaty high-energy solar spectroscopic imager (RHESSI). *Solar Physics*, 210(1), 3–32. <https://doi.org/10.1023/A:1022428818870>
- Lin, Y., Fang, H., Duan, D., Huang, H., Xiao, C., Ren, G., et al. (2025). Enhancing deep learning ionospheric modeling with solar radiation and flare classes. *Solar-Terrestrial Physics*, 130(2), e2024JA033319. <https://doi.org/10.1029/2024JA033319>
- Liu, J., Qian, L., Wang, W., Pham, K., Kong, X., Chen, Y., et al. (2024a). Energy deposition into the ionosphere during a solar flare with extreme-ultraviolet late phase. *Astrophysical Journal*, 963(1), L8. <https://doi.org/10.3847/2041-8213/ad250b>
- Liu, X., Qian, L., Chamberlin, P. C., Chen, Y., Kong, X., Zhang, Q. J. L., et al. (2024b). Data-driven simulation of effects of a solar flare with extreme-ultraviolet late phase on ionospheric electron density. *The Astrophysical Journal*, 974(2), 157. <https://doi.org/10.3847/1538-4357/ad6ddf>
- Machol, J., & Viereck, R. (2016). GOES X-ray sensor (XRS) measurements. Version 1.4.1.
- Mahmoudian, A., Ghorbali, F., & Vazifehkhah Hafteh, M. (2025). First global machine learning model to predict the rate of TEC Index (ROTI) response to X-class solar flares. *Space Weather*, 23(3), e2024SW004247. <https://doi.org/10.1029/2024SW004247>
- Mitra, A. P. (1974). *Ionospheric effects of solar flares*. Astrophysics and Space Science Library, (Vol. 46). <https://doi.org/10.1007/978-94-010-2231-6>
- Müller, D., Cyr, O. C. S., Zouganelis, I., Gilbert, H. R., Marsden, R., Nieves-Chinchilla, T., et al. (2020). The solar Orbiter mission. *Astronomy and Astrophysics*, 642, A1. <https://doi.org/10.1051/0004-6361/202038467>
- Nishimoto, S., Watanabe, K., Jin, H., Kawai, T., Imada, S., Kawate, T., et al. (2023). Statistical analysis for EUV dynamic spectra and their impact on the ionosphere during solar flares. *Earth Planets and Space*, 75(1), 30. <https://doi.org/10.1186/s40623-023-01788-6>
- O'Hare, A. N., Bekker, S., Greatorex, H. J., & Milligan, R. O. (2025). Investigating a characteristic time lag in the ionospheric F-region's response to solar flares. *Atmosphere*, 16(8), 937. <https://doi.org/10.3390/atmos16080937>
- Orni, S., Veronig, A. M., & Dissauer, K. (2025). Extreme ultraviolet late-phase flares as observed by EVE and AIA on board the Solar Dynamics Observatory. *Astronomy and Astrophysics*, 702, A49. <https://doi.org/10.1051/0004-6361/202556242>
- Pesnell, W. D., Thompson, B. J., & Chamberlin, P. C. (2012). The Solar dynamics observatory (SDO). *Solar Physics*, 275(1–2), 3–15. <https://doi.org/10.1007/s11207-011-9841-3>

- Ryakhovsky, I. A., Poklad, Y. V., Gavrilov, B. G., & Bekker, S. Z. (2024). Estimation of the ionospheric D-region ionization caused by X-class solar flares based on VLF observations. *JGR: Space Physics*, 129(6), e2023JA031532. <https://doi.org/10.1029/2023JA031532>
- Sapunova, A., Ryakhovsky, I., Poklad, Y., Gavrilov, B., Lobanova, V., Ermak, V., & Kozakova, E. (2026). Ionospheric and radiophysical effects of the polar region according to the data of the research vessel professor molchanov arctic expedition. *Advances in Space Research*, 77(4), 5196–5205. <https://doi.org/10.1016/j.asr.2025.12.012>
- Snow, M., McClintock, W. E., Crotser, D., & Eparvier, F. G. (2009). EUVS-C: The measurement of the magnesium II index for GOES-R EXIS. In S. Fineschi & J. A. Fennelly (Eds.), *Solar physics and space weather instrumentation iii* (Vol. 7438). <https://doi.org/10.1117/12.828566>
- Solomon, S. C., & Qian, L. (2005). Solar extreme-ultraviolet irradiance for general circulation models. *Journal of Geophysical Research (Space Physics)*, 110(A10), A10306. <https://doi.org/10.1029/2005JA011160>
- The SunPy Community, Barnes, W. T., Bobra, M. G., Christe, S. D., Freeland, S. L., Hayes, L. A., et al. (2020). The SunPy Project: Open source development and status of the version 1.0 core package. *The Astrophysical Journal*, 890(1), 68. <https://doi.org/10.3847/1538-4357/ab4f7a>
- Watanabe, K., Jin, H., Nishimoto, S., Imada, S., Kawai, T., Kawate, T., et al. (2021). Model-based reproduction and validation of the total spectra of a solar flare and their impact on the global environment at the X9.3 event of September 6, 2017. *Earth Planets and Space*, 73(1), 96. <https://doi.org/10.1186/s40623-021-01376-6>
- Woods, T. N. (2014). Extreme ultraviolet late-phase flares: Before and during the solar dynamics observatory Mission. *Solar Physics*, 289(9), 3391–3401. <https://doi.org/10.1007/s11207-014-0483-0>
- Woods, T. N., Eparvier, F. G., Hock, R., Jones, A. R., Woodraska, D., Judge, D., et al. (2012). Extreme ultraviolet variability experiment (EVE) on the Solar Dynamics Observatory (SDO): Overview of science objectives, instrument design, data products, and model developments. *Solar Physics*, 275(1–2), 115–143. <https://doi.org/10.1007/s11207-009-9487-6>
- Woods, T. N., Hock, R., Eparvier, F., Jones, A. R., Chamberlin, P. C., Klimchuk, J. A., et al. (2011). New solar extreme-ultraviolet irradiance observations during flares. *The Astrophysical Journal*, 739(2), 59. <https://doi.org/10.1088/0004-637X/739/2/59>
- Xiong, B., Wan, W., Liu, L., Withers, P., Zhao, B., Ning, B., et al. (2011). Ionospheric response to the X-class solar flare on 7 September 2005. *Journal of Geophysical Research (Space Physics)*, 116(A11), A11317. <https://doi.org/10.1029/2011JA016961>
- Yasyukevich, Y., Astafyeva, E., Padokhin, A., Ivanova, V., Syrovatskii, S., & Podlesnyi, A. (2018). The 6 September 2017 X-Class solar flares and their impacts on the ionosphere, GNSS, and HF radio wave propagation. *Space Weather*, 16(8), 1013–1027. <https://doi.org/10.1029/2018W001932>